Observability and Monitoring Standard

Document ID	ENG-010
Department	Engineering
Document type	Standard
Owner	Head of Site Reliability Engineering
Version	3.2
Status	APPROVED
Effective / last reviewed	2025-09-03
Audience	Engineering staff
Related documents	ENG-004, SEC-015, ENG-014

Summary. Defines how Northstar Systems engineering teams instrument, monitor and alert on production services in NorthWatch, including telemetry retention, golden signals dashboards, alert quality rules, paging criteria, SLO targets and review cadence for Tier-1, Tier-2 and Tier-3 services.

SYNTHETIC DOCUMENT - Northstar Systems is a fictional company. This document exists for development, testing and demonstration only and is not real legal, employment, security or compliance advice.

1. Purpose and Scope

This standard sets the minimum observability and monitoring requirements for every production service owned by Northstar Systems Private Limited engineering teams. Its goals are to detect customer-impacting problems before customers report them, to give on-call engineers the telemetry needed to diagnose problems quickly, and to keep alert volume low enough that every page is taken seriously.

The standard applies to all services running in the Mumbai primary region and the Hyderabad disaster recovery (DR) region, to the Northstar Nexus Platform, Nexus Connect and Nexus Insight components, and to internal platforms that engineering operates. It applies to engineers, engineering managers, Site Reliability Engineering (SRE) and the on-call rotation. Security-specific monitoring (SIEM detections, audit log review) is governed by SEC-015 and is referenced here only where the two overlap.

All telemetry is collected, stored, searched and alerted on in **NorthWatch**, the monitoring, alerting, log search and SIEM platform. Teams must not run parallel, unmanaged monitoring stacks for production services. Where a team believes NorthWatch cannot meet a need, it should raise the gap with SRE before building a workaround.

2. Definitions

- **Telemetry**: the three signal types collected from a service, namely logs, metrics and traces.
- **Golden signals**: latency, traffic, errors and saturation. Every Tier-1 service must have a dashboard for these four signals.
- **SLI (service level indicator)**: a measured quantity that describes service health, for example the proportion of successful requests.
- **SLO (service level objective)**: the internal target for an SLI over a rolling period. SLOs are internal targets and are different from the uptime commitments made to customers in contracts.

- **Error budget**: the amount of unreliability an SLO allows. A 99.95% SLO over 30 days allows roughly 21.6 minutes of unavailability.
- **Symptom-based alert**: an alert on something a customer or a customer-facing function would notice, such as failed requests or slow responses.
- **Cause-based alert**: an alert on an internal condition that may or may not affect customers, such as high CPU on a node.
- **Service tier**: Tier-1 (customer-facing, revenue critical), Tier-2 (internal critical) or Tier-3 (non-critical), as defined in ENG-004.
- **Runbook**: a document, linked from the alert, that tells the responder what the alert means and what to do first.
- **Page**: an alert delivered to the on-call engineer by phone or push notification that requires acknowledgement.

3. Roles and Responsibilities

- **Service owner (engineering team lead)**: ensures the service meets this standard, owns its dashboards, alerts and runbooks, and reviews alert quality monthly.
- **Head of Site Reliability Engineering**: owns this standard, owns the NorthWatch platform configuration and approves exceptions.
- **On-call engineer**: acknowledges pages, follows the runbook, escalates when needed and records alert feedback after each shift.
- **Engineering manager**: receives auto-escalated pages, approves alert-tuning changes for their teams and signs off the quarterly alert review.
- **Security Operations**: owns security detections in NorthWatch. Security alerts follow SEC-015 and the security incident process, not the service paging rules in this document.

4. Telemetry Requirements

4.1 Logs

Every service must emit structured logs in JSON with at minimum a timestamp in UTC, service name, environment, log level, request or trace identifier and a message. Logs must not contain secrets, passwords, tokens, full card numbers or unmasked personal data. Where a log line must reference a customer, it should use the tenant ID rather than names or contact details.

Retention of application logs in NorthWatch is as follows.

Stage	Retention	Notes
Hot (searchable instantly)	30 days	Used for live debugging and incident work
Archived (restore on request)	12 months	Restored through a NorthDesk request to SRE
Security audit logs	24 months	Tamper-evident storage; governed by SEC-015

After 12 months in the archive, application logs must be deleted. A legal hold overrides this rule and is applied by Legal through the Security Desk.

4.2 Metrics

Every service must expose metrics for the four golden signals and for its key dependencies (databases, queues, downstream APIs). Metrics are retained for 13 months so that teams can

compare a given month against the same month in the previous year. Metric names should follow the common naming convention published by SRE, and labels must have bounded cardinality. A label that can take unlimited values, such as a user ID or a request ID, must not be used on a metric.

4.3 Traces

Tier-1 services must emit distributed traces for all inbound requests, with sampling that retains every errored trace and every trace slower than the service latency objective. Traces are retained for 15 days. Trace context must propagate across service boundaries, including asynchronous queue hops, so that a single customer request can be followed end to end. Tier-2 services should emit traces; Tier-3 services may.

4.4 Summary of retention

Signal	Retention
Logs, hot	30 days
Logs, archived	12 months
Metrics	13 months
Traces	15 days
Security audit logs	24 months

Retention for backups and recovery data is covered in ENG-014 and is separate from the telemetry retention above.

5. Dashboards

Every Tier-1 service must have a golden signals dashboard in NorthWatch before it is released to production. The dashboard must show, at a minimum, the following.

1. **Latency**: p50, p95 and p99 request latency, split by successful and failed requests.
2. **Traffic**: requests per second, and for Nexus workloads the number of Pipeline runs and Webhook deliveries per minute.
3. **Errors**: error rate as a percentage of requests, split by error class (4xx and 5xx), and for Nexus services the volume of NX-series error codes.
4. **Saturation**: CPU, memory, disk, connection pool usage and queue depth against their limits.

The dashboard should also show the current SLO attainment and the remaining error budget for the month, the version currently deployed with the canary and rollout state, and the health of the top three dependencies. Dashboards for Tier-2 services should show golden signals; Tier-3 services need only a basic health view.

Dashboards must be created from the shared dashboard templates maintained by SRE so that on-call engineers see a familiar layout when moving between services. Changes to a template are announced in the engineering channel at least five business days before they take effect.

6. Alerting Rules

6.1 Quality requirements for every alert

An alert may be enabled in production only when all of the following are true.

- It has a clear name that states the symptom, for example "Orders API error rate above 2%".
- It has an owner team recorded in NorthWatch.
- It links to a runbook. An alert must have a runbook link; alerts without one must be disabled or rejected in review.
- It has a severity that matches the incident severity model in ENG-004 (SEV1 to SEV4).
- It has a threshold and duration chosen from observed data, not guessed.
- It is tested in a non-production environment or fired manually once to prove the notification path.

6.2 What may page

Paging alerts only for customer-impacting symptoms. A page wakes a person, so it must meet a strict test: a customer is affected now, or will be affected soon unless a human acts, and the action cannot wait until business hours. Typical paging alerts are sustained elevated error rate, sustained latency above objective, failed synthetic checks against public endpoints, and fast burn of the error budget.

Cause-based alerts, such as high CPU, low disk space with days of runway, certificate expiry weeks away or a single node restart, must not page. They should create a ticket in NorthDesk or post to the team channel during business hours.

6.3 Runbook content

A runbook linked from an alert must contain: what the alert means in one or two sentences, the likely customer impact, the first three diagnostic checks with the NorthWatch queries or dashboard links to use, the safe first mitigation (for example rollback, scale out, failover), when to escalate and to whom, and the date it was last reviewed. Runbooks must be reviewed at least every 12 months and after any incident in which they were found to be wrong or missing.

6.4 Severity mapping and acknowledgement

Severity	Typical alert condition	Delivery	Acknowledge within
SEV1	Complete outage, data loss or confirmed customer-affecting security breach	Page, 24x7	5 minutes
SEV2	Major degradation of a key feature or multiple customers	Page, 24x7	15 minutes
SEV3	Minor degradation with a workaround	Ticket and channel	4 business hours
SEV4	Cosmetic or low impact	Ticket	Next business day

Separately from incident severity, the on-call page itself must be acknowledged within 5 minutes. If it is not, it auto-escalates to the secondary engineer, and then to the engineering manager after another 10 minutes. This paging chain is defined in ENG-004 and is configured in NorthWatch by SRE.

6.5 Automatic rollback signals

During a release rollout, the deployment pipeline watches the same metrics that feed alerts. The automatic rollback trigger is an error rate above 2% or p95 latency above 1.5x baseline for 5 consecutive minutes. Teams must configure their service dashboards and alert rules so that this trigger uses the same error-rate and latency definitions as their paging alerts, otherwise the rollout

and the on-call engineer may disagree about whether the service is healthy.

7. Service Level Objectives

Every Tier-1 and Tier-2 service must have at least one availability SLO and one latency SLO documented in its service record. The internal uptime SLO targets are set out below.

Tier	Availability SLO	Approx. allowed downtime per 30 days
Tier-1	99.95%	about 21.6 minutes
Tier-2	99.9%	about 43.2 minutes
Tier-3	No mandated target	Team may define its own

These are internal targets and must not be quoted to customers. Customer commitments (Standard 99.5%, Premier 99.9%, Enterprise 99.95% monthly uptime) are contractual and are handled by Customer Success; the internal Tier-1 SLO is deliberately aligned with the highest customer commitment so that engineering notices a problem before a service credit becomes payable.

7.1 Error budget policy

- When more than 50% of the monthly error budget has been consumed before mid-month, the service owner must review the causes in the next team meeting.
- When the error budget is exhausted, the team must pause non-essential feature releases for the service and prioritise reliability work until the budget recovers. The Head of SRE and the service owner may agree a documented exception for a security fix or a contractual commitment.
- Error budget status for Tier-1 services is reported in the monthly engineering operations review.

7.2 Measuring availability

Availability is measured from the load balancer or API gateway (for Nexus Connect, from the gateway) as the proportion of valid requests that return a non-5xx response within the service latency objective. Planned maintenance announced to customers at least 7 days ahead is excluded from customer commitments but is still visible on the internal dashboard so that teams do not hide reliability problems behind maintenance windows.

8. Synthetic and Dependency Monitoring

Tier-1 services must have synthetic checks that exercise a representative customer journey every minute from at least two locations outside the production network. For the Nexus Platform this includes authenticating with a test API token, running a trivial Pipeline in a dedicated monitoring Workspace, and confirming that a Webhook delivery is received. Synthetic check failures that persist for 3 consecutive runs are paging alerts.

Teams must monitor the health of critical third-party and internal dependencies. Dependency alerts should be symptom-based where possible: alert on the rate of failed calls to the dependency rather than on the dependency's own status page. Where a dependency issue is customer-visible, the incident process in ENG-004 applies and the Communications Lead updates Northstar Status within 30 minutes of declaring a SEV1 or SEV2.

9. Alert Hygiene and Review

9.1 Monthly alert review

Each service owner must review every alert that fired in the previous month. For every page, the team must record one of four outcomes: actionable and correct, actionable but noisy, not actionable, or missed (a problem occurred without an alert). Alerts that were not actionable must be tuned, downgraded to a ticket or removed within 10 business days.

9.2 Noise targets

An on-call engineer should receive no more than two pages per 12-hour shift on average, and no more than one page per night-time period (22:00 to 06:00 IST) on average over a rotation week. If a team's rotation exceeds these figures for two consecutive rotations, the engineering manager must present a reduction plan to the Head of SRE. Duplicate alerts for the same cause must be grouped so that one problem creates one page.

9.3 Quarterly alert audit

SRE audits a sample of alerts every quarter for compliance with section 6.1. Findings are tracked in NorthDesk with owners and due dates. Services with repeated findings may lose their Tier-1 release approval until the findings are closed.

9.4 Post-incident feedback

Every blameless post-incident review for SEV1 and SEV2 incidents (completed within 5 business days of resolution) must answer three observability questions: did we detect the problem before customers did, did the right alert fire with a usable runbook, and was the telemetry sufficient to find the cause. Gaps become action items in NorthDesk with owners and due dates.

10. Onboarding a New Service

A new Tier-1 service must complete the following steps before its first production release is approved by the Change Advisory Board.

1 Register the service in the service catalogue with its tier, owner team and on-call rotation.
2 Instrument logs, metrics and traces following section 4 and verify they appear in NorthWatch in the staging environment.
3 Create the golden signals dashboard from the template and confirm that it shows data from staging load.
4 Define SLOs and create at least one symptom-based paging alert, with a runbook, for availability and for latency.
5 Add a synthetic check for the primary customer journey.
6 Run a game-day exercise: break the service in staging and confirm that the alert fires, the page is delivered, and the runbook leads to a fix.
7 Record the results in the NorthDesk change ticket so that the CAB can see the evidence.

Tier-2 services follow the same steps except that traces and synthetic checks are recommended rather than mandatory. Tier-3 services need logs, basic metrics and a ticket-based alert for failure.

11. Worked Examples

Example 1: noisy CPU alert. The Billing Reconciliation service (Tier-2) pages the on-call engineer whenever node CPU exceeds 85% for 5 minutes. Over the last month it fired 19 times and nobody took action, because requests stayed fast. Under section 6.2 this is a cause-based alert and must

not page. The team converts it to a NorthDesk ticket alert and adds a symptom alert for p95 latency above objective for 10 minutes. The next monthly review shows zero non-actionable pages.

Example 2: missing runbook. A developer in the Nexus Connect team adds an alert for connector timeout errors (NX-2004) above 5% of calls and wants it live before a Thursday release. The reviewer rejects it because the runbook link field is empty. The developer writes the runbook in 40 minutes: it explains that the default connector timeout is 30 seconds, tells the responder to check endpoint latency in the dashboard and to raise the timeout up to 120 seconds only for the affected connector, and says to open a P2 or higher ticket if the pattern is not connector-specific. The alert is approved the same afternoon.

Example 3: error budget exhausted. The Tier-1 Workflow Scheduler service had a 40-minute partial outage on the 9th of the month. Its monthly allowance at 99.95% is about 21.6 minutes, so the error budget is exhausted. The service owner pauses two non-essential feature releases scheduled for the Tuesday release train, and the team spends the following sprint on the connection-pool fix identified in the post-incident review. The security patch due the same week is released with the Head of SRE's documented exception.

Example 4: log retention request. A support engineer needs logs from 7 months ago to investigate a customer complaint. Those logs are past the 30-day hot window but inside the 12-month archive, so the engineer raises a NorthDesk request to SRE for restoration. If the same request concerned logs 14 months old, it could not be fulfilled because application logs are deleted after the 12-month archive period unless a legal hold applies.

12. Exceptions and Escalation

Exceptions to this standard, for example a Tier-1 service that cannot yet emit traces, must be requested in NorthDesk by the service owner, approved by the Head of SRE and given an expiry date of no more than 90 days. Repeated renewal of the same exception requires the VP Engineering's approval. A service that has no approved exception and does not meet the Tier-1 minimum requirements cannot receive a Tier-1 approval from the Change Advisory Board.

Questions about NorthWatch configuration should go to SRE through NorthDesk. Suspected security events seen in monitoring must be reported immediately to the Security Desk (ext. 5555 or security@northstar-systems.example) and are handled under the security incident response process rather than as a service incident alone. Regulatory expectations about log retention are illustrative and to be verified by Legal before external use.

13. Frequently Asked Questions

Can I page on high CPU? No. Paging alerts only for customer-impacting symptoms. Use a ticket-based alert for CPU unless CPU saturation is already causing slow or failed requests, in which case alert on the latency or error symptom.

How long can I search logs without a request? Logs are searchable for 30 days in the hot tier. From day 31 to 12 months they must be restored from the archive on request.

Do Tier-3 services need dashboards? They need a basic health view and logs, but the full golden signals dashboard is mandatory only for Tier-1 and recommended for Tier-2.

Who decides whether an alert is a SEV1 or a SEV2? The on-call engineer proposes the severity using the alert definition and the incident severity model in ENG-004. The Incident Commander

confirms or changes it.

Are SLO targets the same as customer uptime commitments? No. SLOs are internal engineering targets. Customer uptime commitments and service credits are contractual and managed by Customer Success.

14. Related Documents

- ENG-004 defines incident severities, the on-call rotation and the paging escalation chain that this standard feeds.
- SEC-015 governs security monitoring, audit log retention and review of privileged actions in NorthWatch.
- ENG-014 covers backup, restore testing and disaster recovery, which rely on the monitoring set out here to verify that backups complete.

15. Revision History

Version	Date	Change
3.0	2025-02-10	Added SLO and error budget policy
3.1	2025-06-17	Added synthetic monitoring and alert hygiene targets
3.2	2025-09-03	Clarified retention periods and paging criteria